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Landslide monitoring for risk mitigation by using corner reflector and satellite SAR interferometry: The large landslide of Carlantino (Italy)

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ABSTRACT

The use of satellite Synthetic Aperture Radar Interferometry (InSAR) for monitoring ground instability due to landslide events, although advantageous over large spatial scales, still poses challenges related to the recurrently complex kinematics of the phenomena or to the unfavorable settings of the examined areas with respect to steep topography and vegetated land cover.
This paper presents results obtained by using Multi-temporal InSAR techniques with high resolution TerraSARX (TSX) data for monitoring the Carlantino landslide, located in the Daunian Subapennine (Apulia region, southern Italy) on a slope overlooking a water reservoir, and subjected to several investigations and consolidation works. The targets detected by using Persistent Scatterer Interferometry (PSI) correspond to urban structures or peri-urban walls and guard rails, while the landslide body is almost completely devoid of stable targets, due to the widespread vegetation and variable land cover. To allow stability monitoring, a network of six Corner Reflectors (CR) was designed and deployed over the landslide test site. The TSX images were analyzed by using both the PSI processing and a procedure, based on the double difference analysis of InSAR phase values on the CR pixels. Despite residual noise and the loss of 2 CRs due to vandalism, the processing allowed verifying the stability of the upper and central part of the landslide body, and relating indirectly the movements at the toe of the landslide to the water level fluctuations of the reservoir. Finally, this experiment suggests some recommendations and guidelines in planning CR deployment in complex landslide sites.

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1. Introduction

Landslide instability can be attributed, at local and regional scale, to several predisposing and triggering factors, both environmental, such as geolithology, geomorphology, hydrogeology, seismicity, climate etc., and anthropic, such as land use, changes of slope geometry, or load distribution variations (excavations, fills, building activity, etc.).

The control and managing of the instability is necessary to prevent disasters, especially in territories with high settlement density, or where landslides border the urban area, threatening structures and infrastructures (Dai et al., 2002; Sassa et al., 2005; Masure and Lutoff, 2008; Di Martire et al., 2012; Pellicani et al., 2014a).

The care and managing of territory, increasingly in disuse, can be useful and effective for controlling instability phenomena of reduced areal extent. Generally, the risk level increases in proportion to the size of landslides. Risk reduction for wide landslides cannot be achieved only through stabilization measures, due to their high costs. Risk mitigation in territories affected by diffused instability phenomena can be realized mainly through adequate monitoring and early warning systems (Wilson et al., 1993; Sassa et al., 2005; Capparelli and Tiranti, 2010; Liao et al., 2010; Yin et al., 2010; Intrieri et al., 2012; Thiebes, 2012; Mandal and Maiti, 2015). In these contexts, monitoring of landslides can be considered as the most suitable measure for risk mitigation, especially in terms of reduction of loss of human life and assets.

A risk mitigation approach based on hazard management should take into account the extent of instability phenomena, their state of activity, the failure kinematics (fast or slow, preceded by long preparation time characterized by deformations and millimetric displacements), localization and typology of elements at risk and possibility of designing an adequate early-warning system. It is important to highlight the difference between measurement activities and monitoring, where the second is expected to be the first part of a process, which continues with analysis and decision making.

Among different monitoring techniques, spaceborne SAR Interferometry (InSAR) is attractive for landslide instability monitoring because of its capability to provide regional scale coverage and, under favourable conditions, spatially dense information on ground surface displacements, even of limited size (Rott et al., 1999; Hilley et al., 2004; Strozzi et al., 2005; Bovenga et al., 2006; Casagli et al., 2010; Prati et

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al., 2010; Lauknes et al., 2010; Bianchini et al., 2012; Bovenga et al., 2012a; Herrera et al., 2013; Bovenga et al., 2013). In particular, advanced multi-temporal InSAR techniques, such as Persistent Scatterer Interferometry (PSI) are able to detect and monitor, with millimetric precision, displacements occurring on selected radar targets, or Persistent Scatterers (PS) (Ferretti et al., 2001; Colesanti and Wasowski, 2006; Farina et al., 2006), exhibiting stable radar backscattering properties. PS targets correspond mainly to manmade structures or to rock outcrops; their spatial density depends on the ground coverage, being maximum over urban areas. The application of satellite multi-temporal InSAR analysis to ground instability monitoring has been well assessed in the last years for several geophysical phenomena (Wegmüller et al., 2010; Sansosti et al., 2010; Hooper et al., 2012). Nevertheless, the systematic application of these techniques to landslides still poses challenges related to the complex kinematics of the phenomenon, as well as to the unfavorable settings of the majority of examined areas (e.g. steep topography, incoherent land cover). A review of different factors that influence the applicability of multi-temporal InSAR techniques to landslide investigations is presented in Wasowski and Bovenga (2014).

By increasing the number of measurable targets, high-resolution SAR sensors, such as TerraSAR-X (TSX), COSMO-SkyMed (CSK) or ultra-fine-mode Radarsat-2, make InSAR applications very promising for monitoring single man-made structures (e.g. buildings, bridges, railways and highways), as well as areas with low PS density (Bovenga et al., 2012a; Wasowski and Bovenga, 2014).

This study presents the results obtained by using InSAR techniques for monitoring a large landslide, located on the western border of the urban area of Carlantino, a small town belonging to the Daunian Subapennine (Apulia region, Southern Italy). The relevance of this landslide, which extends for nearly 2 km from the hilltop down to the valley, derives from the presence of the built-up area in the upper part, and of a water reservoir, closed by a dam on the Fortore River, at the toe. It is common knowledge that filling and exploitation of dam reservoirs has a negative effect on the stability of the surrounding slopes, and thus leads to intensification of landslide activity (Deng et al., 2005; Xue et al., 2009; Kaczmarek et al., 2015). The operations associated to the dam use produce changes of the water level, which often cause the activation of pre-existing landslides and the trigger of new ones. Many studies were carried out in order to evaluate the stability of reservoir landslides (Du et al., 1996; Wang et al., 2007; Zhang et al., 2013; Xu et al., 2014).

In this study, spaceborne InSAR monitoring was aimed at checking firstly the landslide state of activity (to detect e.g. velocity variations) and, secondly, a possible correlation between movements at the toe and the periodic changes of the reservoir level. To this aim, both C-band ENVISAT and X-band TSX data were processed through PSI. The detected PS targets correspond to urban structures or peri-urban walls and guard rails, while the landslide body is almost completely devoid of stable targets, due to the variable land cover, mainly vegetated, of the landslide body. In order to extend stability monitoring to this area, a network of six Corner Reflectors (CR) was designed and deployed over the landslide test site. In spite of some technical problems due to the harsh conditions of the area, and the loss of 2 CRs due to vandalism, analysis of TSX data acquired after CR deployment through both PSI techniques and processing of point-wise phase values led to reliable conclusions about the above-mentioned stability issues.

2. Study area: the Carlantino landslide

2.1. General description

The monitored landslide is located on the western hillslope of Carlantino, a small town belonging to the western sector of Daunian Subapennine, in Apulia Region (Fig. 1*a*), area affected by widespread landslide phenomena (Puglisi et al., 2005; Pellicani and Spilotro, 2015). The Daunian region is generally characterized by scarce

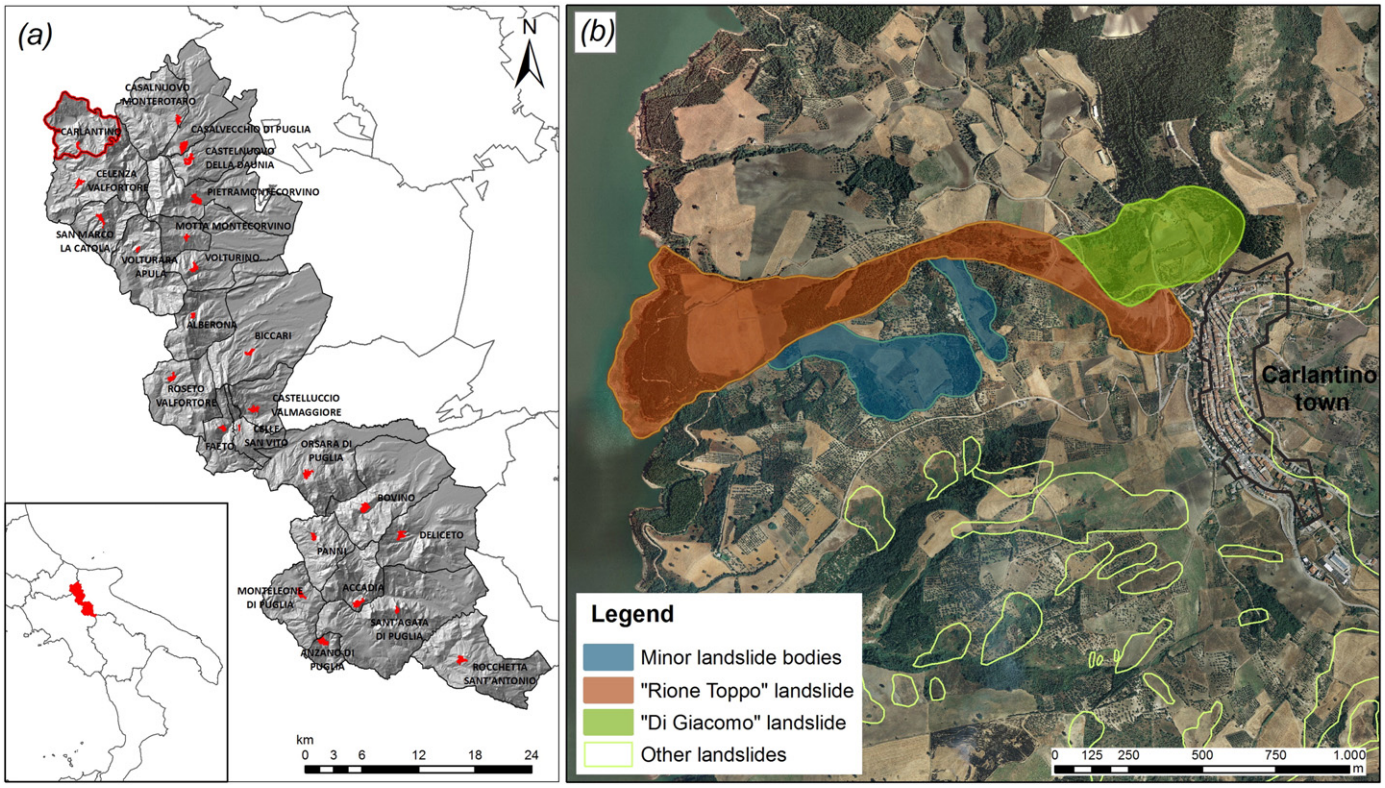


Fig. 1. (a) Western sector of Daunian Subapennine, in Apulia Region, and territory of Carlantino highlighted in red; (b) Landslides affecting the western hillslope of Carlantino, composed by the oldest one, "Di Giacomo" landslide, the main landslide body, "Rione Toppo" landslide, and other minor landslides bodies.

urbanization with hill-top towns scattered across a large agricultural and densely vegetated area, and has experienced landslide problems in the past and more recent years, mainly due to the poor geotechnical properties of the soil (Pellicani et al., 2014b). This geographical area, at the NW edge of the Apulia Region, is a territory characterized by complex terrains under the geological, hydrogeological and geotechnical point of view. Here, in fact, ancient and more recent (Pleistocene) tectonics have put in contact very different formations, by lithology and technical characters, impressing evident fracturing networks, that, especially in clayey soils, make them susceptible to landslide and erosion phenomena (Melidoro, 1971; Zezza et al., 1994). The morphology of the territory surrounding the urban area of Carlantino is affected by different landslide typologies, as a consequence of strong distensive tectonics, affecting both the flyschoid formations and the Plio-Pleistocene sedimentary sequences.

The most important and significant landslide, herein examined, is located at the western border of the urban area of Carlantino. This complex landslide, known as "Rione Toppo" landslide, extends for nearly 2 km from the hilltop down to the valley and is characterized by a main feeding zone (Fig. 1*b*), positioned along a marked tectonic lineation at the upper part of the landslide body, and secondary feeding areas along the sides of the landslide channel, itself developing along a fault plane as right border (Canora et al., 2012). The relevance of the landslide lies in the presence of two risk elements: the town built-up area in the upper part, and the water reservoir on the Fortore River at the toe (Fig. 1*b*). Several investigations were carried out in the last 30 years in order to define the stratigraphic, geotechnical and hydrogeological setting of the area (Daurù et al., 1989; Puglisi et al., 2005), and to collect data useful for both monitoring the landslide activity and designing consolidation works. The field surveys, consisting in geognostic boreholes for reconstructing the stratigraphic profile and extracting undisturbed samples, piezometric and inclinometric measurements, temperature and salinity values, natural radioactivity and other geophysical logs, were used to reconstruct the tectonic and geological framework of the landslide, and to assess the role of groundwater, which acts with different mechanisms at the top, where the feeding area was located, and at the bottom, where the base flow level is constituted by the seasonal reservoir level. From 1988 to 2008, a number of consolidation works were designed and realized with the aim of stabilizing the landslide body. In particular, the remodeling and regularization of the crown and of the debris masses along the landslide channel were carried out through deep and surface drainage systems, and some piling for reinforcing the existing unstable soil.

2.2. Geological and hydrogeological setting

The territory of Carlantino is located in a contact zone between two tectonic units of regional relevance, named Daunia Unit and Fortore Unit, respectively. The contact between the two units is represented by a thrust, recognizable on regional scale, which is aligned with the NW-SE direction (Dazzaro and Rapisardi, 1984; Dazzaro and Rapisardi, 1996).

The Daunia Unit outcrops at the top of the hillslope affected by the landslide and is represented mainly by the Faeto Flysch Formation. The Fortore Unit outcrops largely in the central and terminal part of the slope, and consists of the Varicolored Clays, San Bartolomeo Flysch

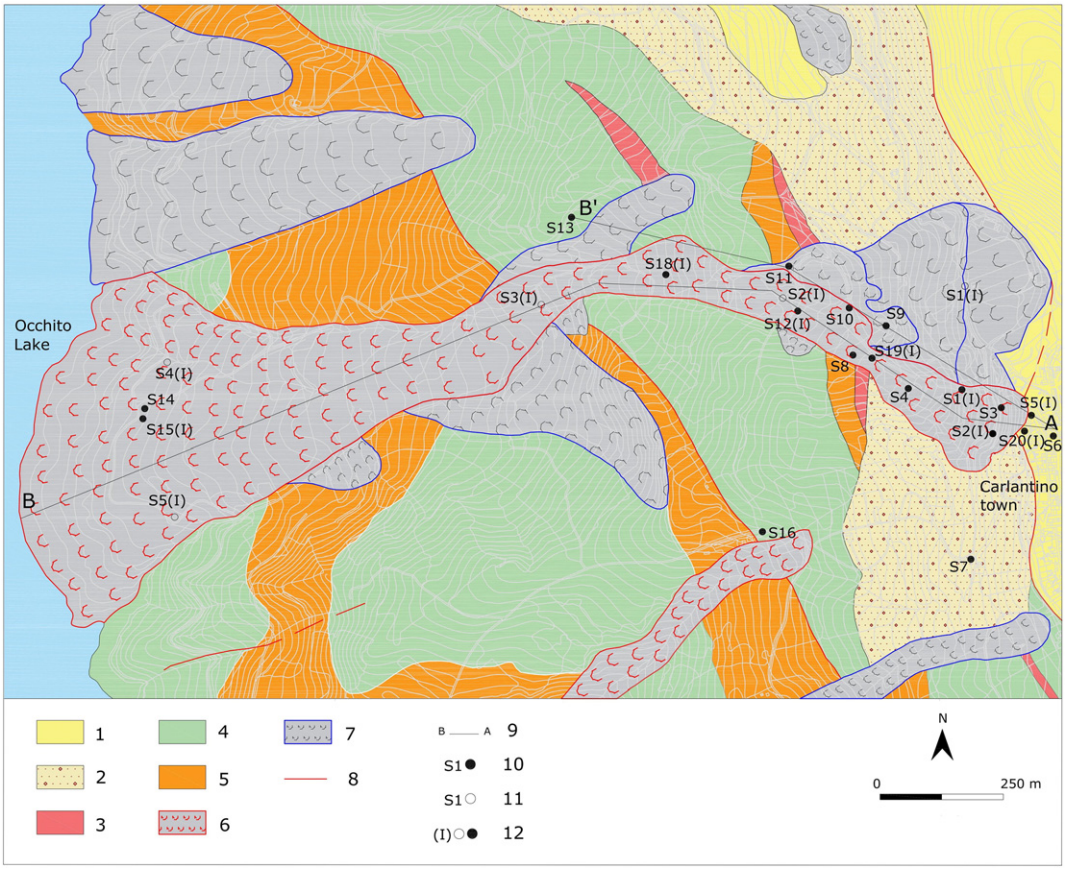


Fig. 2. Geomorphological map of western hillslope of Carlantino (modified from Santagati, 1988) with the indication of the outcropping lithologies. (Legend: 1 – Faeto Flysch Fm., 2 – Debris deposit, 3 - Varicolored Clays Fm., 4 – Toppo Capuana Clayey-Marls Fm., 5 – San Bartolomeo Flysch Fm., 6 – Active landslide, 7 – Inactive landslide, 8 – Fault, 9 – Landslide section, 10 – Surveys carried out in 1993, 11 - Surveys carried out in 2005, 12 – Inclinometers).

and Toppo Capuana Clayey-Marls Formations (Costella and Mezzani, 1964; Dazzaro and Rapisardi, 1987; Patacca and Scandone, 2007).

In particular, at the top of the slope, at the edge of the built-up area, a large outcrop of debris deposit overlaps the calcareous-marly faces of Faeto Flysch and, downstream, involves the clayey-marly terms of Toppo Capuana Formation and the clayey-arenaceous sequences of San Bartolomeo Flysch, with intercalations of scaly clays with chaotic structure belonging to the Varicolored Clays Formation (Fig. 2).

The stratigraphic relationships among the overlapped geological formations are disrupted due to the presence of monoclines, overturned and dipping at NE and affected by faults, which determine the repetition of the series, with the overlap of the Faeto Flysch on the San Bartolomeo Flysch (Fig. 3). A sub-vertical fault, with NW-SE orientation and with associated and conjugated plans, determines alignment of the crowns of both old and recent landslides. Mass movements with lower intensity occurred along the slope, affecting all the cited flysch formations, and feeding the landslide channel with debris of different lithological and granulometric composition.

The hydrogeological framework of the study area is widely influenced by the presence of a wide reservoir at the toe of the landslide, closed by the Occhito dam. The reservoir is characterized by an important evaporating surface, which exerts a moderating action on winter temperature. It generates an attenuation of the climate, more sensitive on the western side of the area, and, at the same time, provides moisture during summer and autumn. The annual rainfall in the Daunian Subapennine region is about 800–850 mm, 60% of which concentrated in autumn-winter months, while the average annual temperature is around 12 °C.

The hydrogeology of the study area is characterized by remarkable complexity, deriving from the overlap of formations and lithologies, and from the general and local structural characteristics of the involved formations, to which discontinuity families are added with persistence from moderate to low.

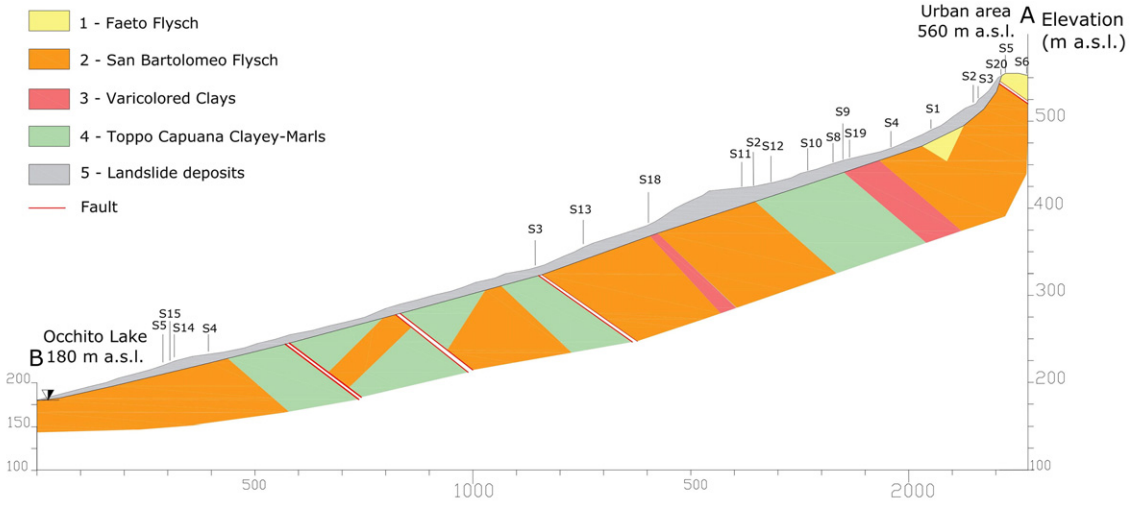
The hydrogeological formations present along the slope, in order of increasing permeability, are the following:

- Varicolored Clays;
- clayey-marly facies of the San Bartolomeo Flysch and the Toppo Capuana clayey marls;
- sandstone and limestone-marly members of the San Bartolomeo Flysch and Faeto Flysch.

The landslide body, in its various parts, was subjected to numerous investigations. More specifically, in relation to the hydrogeological surveys, piezometric level measurements, temperature and natural radioactivity logs were recorded, for reconstructing the water circulation in the different parts of the landslide body.

The landslide can be divided into four morphological sectors, which have different behaviors both in kinematics and hydrogeological terms: the upper part, located around the large scarp generated in 1976, between 540 and 465 m a.s.l.; the middle part, between 465 and 395 m a.s.l.; the lower part, between 395 and 245 m a.s.l.; and the toe, which extends from 245 to 170 m a.s.l., on the ancient bed of the Fortore River, and currently covered by the artificial lake.

In the upper part, the highly fractured calcarenite formations of the Faeto Flysch, confined at the bottom by the clay term of the San Bartolomeo Flysch, determine an aquifer. This feeds, for a stratigraphic convergence, the NE hillslopes on which the town stands, and also, due to transverse fracture systems, the opposite side, saturating and pressurizing the same tectonic discontinuities. In this part, the presence of cold fresh groundwater was recognized, which is representative of feeding and infiltration areas, and of a relatively fast circulation affecting



Fm.	Lithotypes	Gravel (%)	Sand (%)	Silt (%)	Clay (%)	LL (%)	Plasticity Index	Consistency index	Activity
2	Clayey silt	0	18÷22.7	29.4÷46.4	36.5÷52	42.5÷54.8	22.9÷32.5	1÷1.4	0.58÷0.89
4	Silty clay	0	4.2÷27.9	18.4÷44.2	28.7÷62	66.1÷98.5	35.4÷47.1	1.1÷1.4	0.66÷1.45
1-2-4	Calcareous-marly facies	4.2÷28.1	21.4÷35	19.5÷37	23.4÷34.1	25.4÷55.9	13.2÷21	0.5÷0.75	0.5÷1.5
2	Sandy gravel	14.2÷42.6	19.6÷40	13.8÷36.7	11.5÷25.5	32.9÷44.8	6.5÷27.6	0.25÷0.75	0.75÷1.25
3	Silty clay (high plasticity)	0	<10	19÷40	35÷79	72÷154	43÷126	>1	0.79÷2.15
2-4	Sandy clay-debris	0÷20	10÷41.6	18.2÷43.3	23.1÷54.5	36.1÷79.7	17.4÷58.9	>1	0.5÷1.25
2	Sand	0÷15	38÷63	16÷25	16÷25	24÷29	16÷29	1.2÷1.3	0.65÷1.45

Fig. 3. Lithostratigraphical section (Santagati 1988, modified) of the landslide of "Rione Toppo" (A–B in Fig. 2) and grain size and physic-compositional parameters of seven main lithotypes constituting the four geological formations (1 – Faeto Flysch, 2 – San Bartolomeo Flysch, 3 – Varicolored Clays, 4 – Toppo Capuana Clayey-Marls).

a deep strip corresponding to the main tectonic lineation. The profiles of isotherms and isohalines confirmed clearly the presence of cold freshwater, defining the feeding and infiltration areas at the top of the landslide body (Fig. 4*a*–*b*). In the central part, corresponding to the landslide channel, the presence of heterogeneous debris deposits, and the morphological disorder of the surface, define a slower circulation in the direction of maximum slope (Fig. 4*a*–*b*). The accumulation area of the landslide, which extends up to the bed of the Fortore River, currently appears to be partially submerged by the Occhito reservoir, determined by a dam. The groundwater level, in this part of the landslide body, follows, with some delay, the lake level fluctuations. Since 1988, shallow and deep drainage systems have been realized in order to produce a perennial water discharge, which is spilled into a surface drainage channel and is used for irrigation.

2.3. Kinematics and evolution of the landslide

A detailed geomorphological analysis conducted on the wide area between the built-up area of Carlantino and the Fortore valley, where the Occhito lake is located, allowed recognizing different landslide bodies, which converge into a single landslide channel. The geological in-situ surveys allowed also to highlight a main fault structure, aligned in the WNW-ESE direction, which skirts the urban area of Carlantino. Two large landslides were generated from this fault, the "di Giacomo" landslide, in an ancient unknown period, and the "Rione Toppo" landslide in 1976, as well as minor landslide bodies in 1992, 1997 and 2005 (Fig. 1*b*). The main landslide channel is overridden on a second tectonic structure of transcurrent type, from the edge of which small landslide bodies were generated in unknown time (Fidelibus et al., 1991).

The main landslide body has a complex movement, consisting of a roto-translational slide in the upper part, which feeds an earth flow along the landslide channel. The large scarp of the "di Giacomo" landslide could be the first important instability event, which was followed by the other one, described above. It is important to note that not all the debris produced by the landslide have reached the toe. The large-sized debris, such as that produced by the 1976 event, have never gone below 465 m a.s.l., while the clayey debris have determined much more mobile flows.

The geomorphological complexity of the slope, due to the numerous landslides feeding a single landslide channel, does not allow to determine a specific cause of the instability and the lateral reactivation.

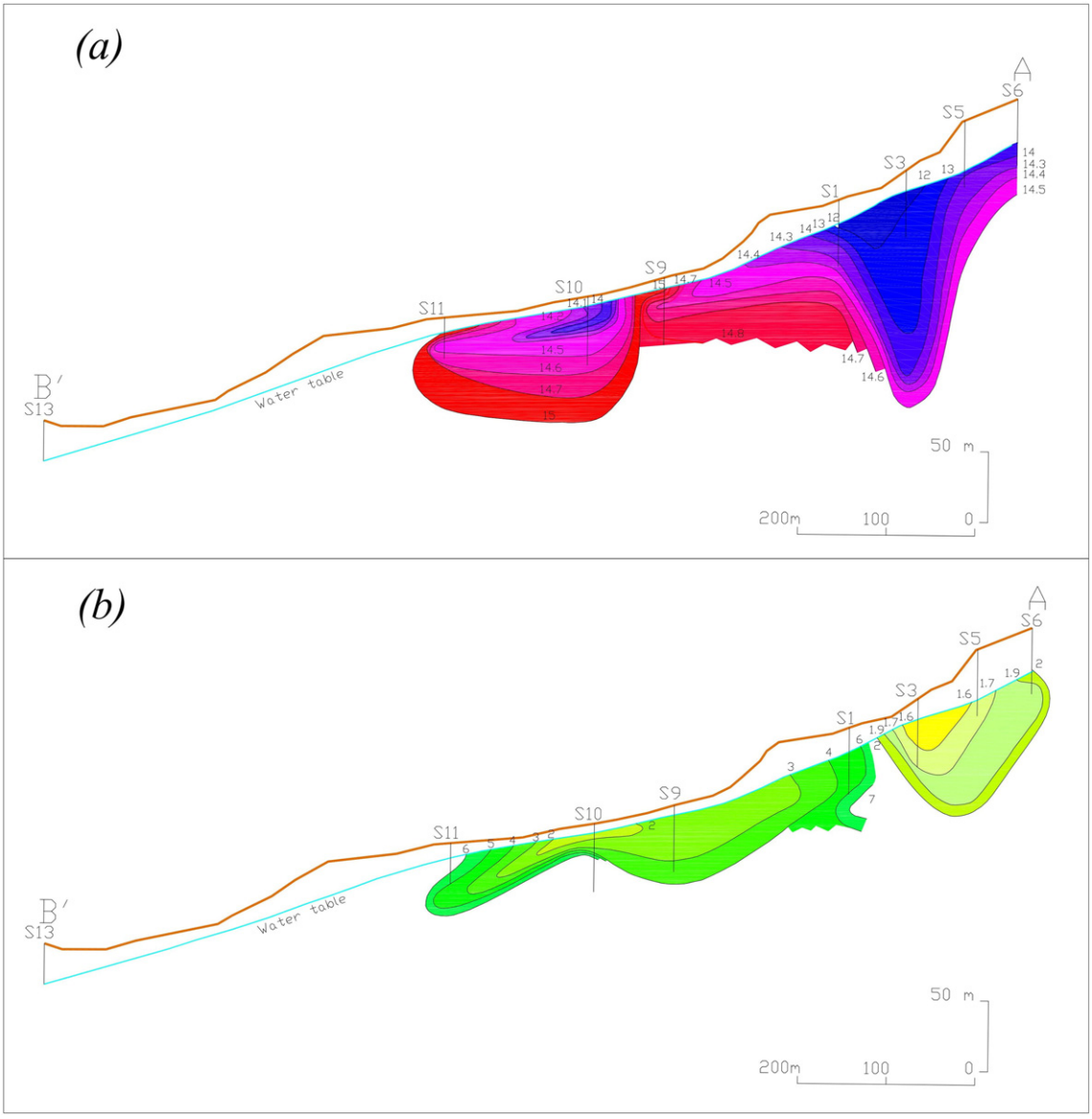


Fig. 4. Trend of isotherms (values expressed in Celsius degree) (a) and isohalines (values expressed in g/l) (b) within the northern part of the landslide (section A–B′ in Fig. 2).

Lithological, stratigraphic and tectonic features, climate, groundwater, geotechnical characteristics of the soils, and seismicity, all interact closely in creating the instability conditions.

The presence of the artificial lake at the toe of the landslide creates also an unfavorable condition for the slope equilibrium. The creation of the lake reduced the erosive activity at the toe of the slope and favored the stability with a moderate silting of the reservoir. However, the hydraulic regimen, seasonally ranging between the maximum and minimum reservoir elevation, induce severe seepage towards the reservoir and adds seepage forces to the stress field. Finally, the interaction between the fresh water of the lake and the interstitial water of clayey soils and the calcareous lithologies contributed to the structural weakening of the soils and to the recurring reactivations of the phenomenon.

The landslide was subjected to different works aimed to stabilize it, through drainage and consolidation systems, and through reconstruction of a vegetated cover at the main scarp and at the top of the landslide channel. In particular, the remodeling and regularization of the crown (Fig. 5*a*–*b*) and of the debris masses along the landslide channel (Fig. 5*c*–*d*) were carried out through the realization of deep and shallow drainage systems, and some piling for reinforcing the existing unstable soil.

Despite the above mentioned consolidation works, it was detected, during inclinometer monitoring performed in 2005, that the toe of the landslide is active and moves with low velocity, such as to cause the breaking of two inclinometer pipes (S4 and S5 in Fig. 2). The lack of recent feeding landslide events (in the last 40 years), allows assuming that the current instability is due to the fluctuations of the lake level. The space-borne InSAR monitoring, described in the following, was carried out in order to assess the landslide current state of activity and possible correlations between movements at the toe and lake fluctuations.

3. Landslide monitoring by spaceborne InSAR

The use of space-borne SAR interferometry in assessing the current state of activity of "Rione Toppo" landslide is aimed at integrating and supporting the traditional in-situ measurements by exploiting the wide coverage and the periodic revisit provided by this remote sensing technique. European Space Agency's (ESA's) ERS-1/2 and ENVISAT SAR data, acquired at C band (5.6 cm wavelength) in stripmap mode (yellow frame in Fig. 6), covering the period 1995–2010 were processed through the SPINUA (Stable Point INterferometry over Un-urbanised Areas) PSI multi-temporal processing technique (Bovenga et al., 2004), detecting several unstable areas (Bovenga et al., 2006).

More recently, a dataset of 40 TSX SAR scenes in total has been acquired at X band (3.1 cm wavelength) in stripmap mode (red frame in Fig. 6), between 2010 and 2013, along ascending orbit, with spatial resolution of 3 m and 2 m in azimuth and range directions, respectively. Thanks to the high resolution of TSX data, the spatial PS density increases up to 40 times with respect to the C-band results, in line with the numbers provided by previous studies (Nutricato et al., 2013).

Nevertheless, the application of PSI techniques for slope instability monitoring still remains challenging in some rural areas, due to the lack of natural coherent targets, as in the case of Carlantino. Fig. 7 shows the spatial distribution and average Line of Sight (LOS) velocity of PSs (marked by colored dots), derived from ASAR ENVISAT (Fig. 7*a*) and TSX (Fig. 7*b*) data, respectively, around the "Rione Toppo" landslide, marked by a yellow polygon. PS targets detected on both C- and X-band data correspond mainly to urban structures or peri-urban walls and guardrails. The PS density from TSX PSI processing increases consistently with respect to C band (about 13 times more TSX PS are found w.r.t. the ENVISAT ASAR results), and several more artificial structures or outcrops can be monitored, which cover part of the top of the landslide (already stabilized) and the road bordering another unstable slope on the eastern side of the Carlantino urban area, evidenced by a magenta filled polygon in Fig. 7(*b*). Nevertheless, the main landslide body clearly appears lacking stable targets, mainly due to vegetation and variable land cover. In such cases, use of artificial reflectors (AR) can be attempted in place of natural stable scatterers, to extend the application of stability monitoring through spaceborne InSAR to difficult areas (Froese et al., 2008; Wenxue et al., 2010; Riedmann et al., 2011, Strozzi et al., 2013). In this particular case, the availability of high resolution, short-wavelength X-band satellite data makes this solution attractive, since artificial reflector of limited size can be used.

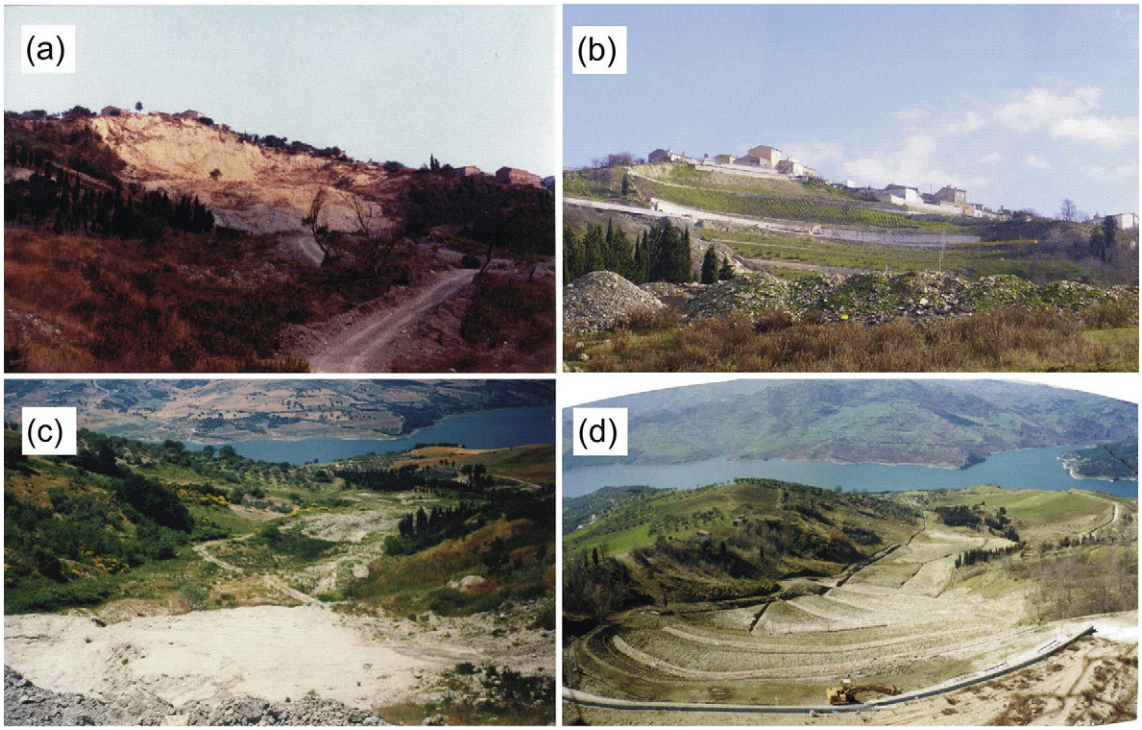


Fig. 5. Photos relative to the landslide of the "Rione Toppo" took in 1978 after landslide movement (a)/(c) and 2006 after consolidation works (b)/(d).

3.1. Artificial reflector design and deployment

The advantage of using artificial reflectors (ARs) instead of natural stable scatterers (PS) is that, in principle, it is possible to design a network of ARs according to geodetic requirements. In particular, ARs can be positioned to potentially cover the whole area of interest, and with arbitrary spatial density. Also, their relative positioning could be properly set to guarantee i) a good sensitivity to the relative motions within the landslide, and ii) an effective signal analysis, concerning in particular the suppression of the atmospheric component of the InSAR phase signal. Furthermore, spatial redundancy could be used to guarantee reliability and precision of the measurements. In case of InSAR analyses, the deformation temporal sampling is the only factor that still remains dependent exclusively on the satellite revisit time.

However, the design of an AR network needs to match specific requirements. The AR has to be designed and deployed so that it results visible by the satellite. This leads to requirements in terms of size, positioning and pointing direction, to avoid obstacles along the signal propagation path to the AR and back, and to allow the AR to be detected with sufficient intensity by the sensor. In particular, shadow and layover areas have to be avoided (Fig. 8*b*). A further requirement concerns the accessibility of the location on the ground, which cannot be taken for granted in case of sites affected by slope instability (steep topography, brittle soil, thick vegetation).

In the present case, Trihedral Corner Reflectors (CRs) were used, each consisting of three triangular metal panels welded perpendicularly to each other to form a trihedral shape, which ensures that the radar signal is scattered almost entirely back to the sensor within a quite wide range of incidence angles (about 40° at 3 dB level). The CRs were deployed and aligned according to an acquisition geometry of ascending orbits, with incident angle $\theta = 35°$ (Fig. 8*a*). Although for slopes facing westward the descending acquisitions would have provided the best sensitivity to the downward movement, we were forced to use the ascending mode, because a large dataset of scenes in this geometry was already available in the framework of another project. Moreover, the ascending geometry allowed to select a few more locations for CR deployment, which would have been not visible in the descending images, due to the presence of trees and other tall objects nearby. The final InSAR data stack consists thus of 40 scenes, 14 acquired before, and 26 after CR deployment. This dataset is suitable for PSI processing, and provided support to the CR-specific InSAR processing detailed in the following.

In Fig. 9, a scheme of the CRs structure and ground fixing is shown, together with a picture of one sample CR deployed in the field. The CR side length is 79 cm, leading to a theoretical normalised Radar Cross Section (RCS) of about 18 dB. Cheap solutions both in the design (welded metal plates) and in the manufacturing (galvanized aluminum) were adopted, which probably led to a certain decrease of the RCS, due to either not perfectly orthogonal or deformed metal plates (Curlander and McDonrough, 1991). Nevertheless, this still allowed the CR to be visible in the SAR imagery. Robust deployment required the construction of concrete basements about 30×30 cm^2 wide and

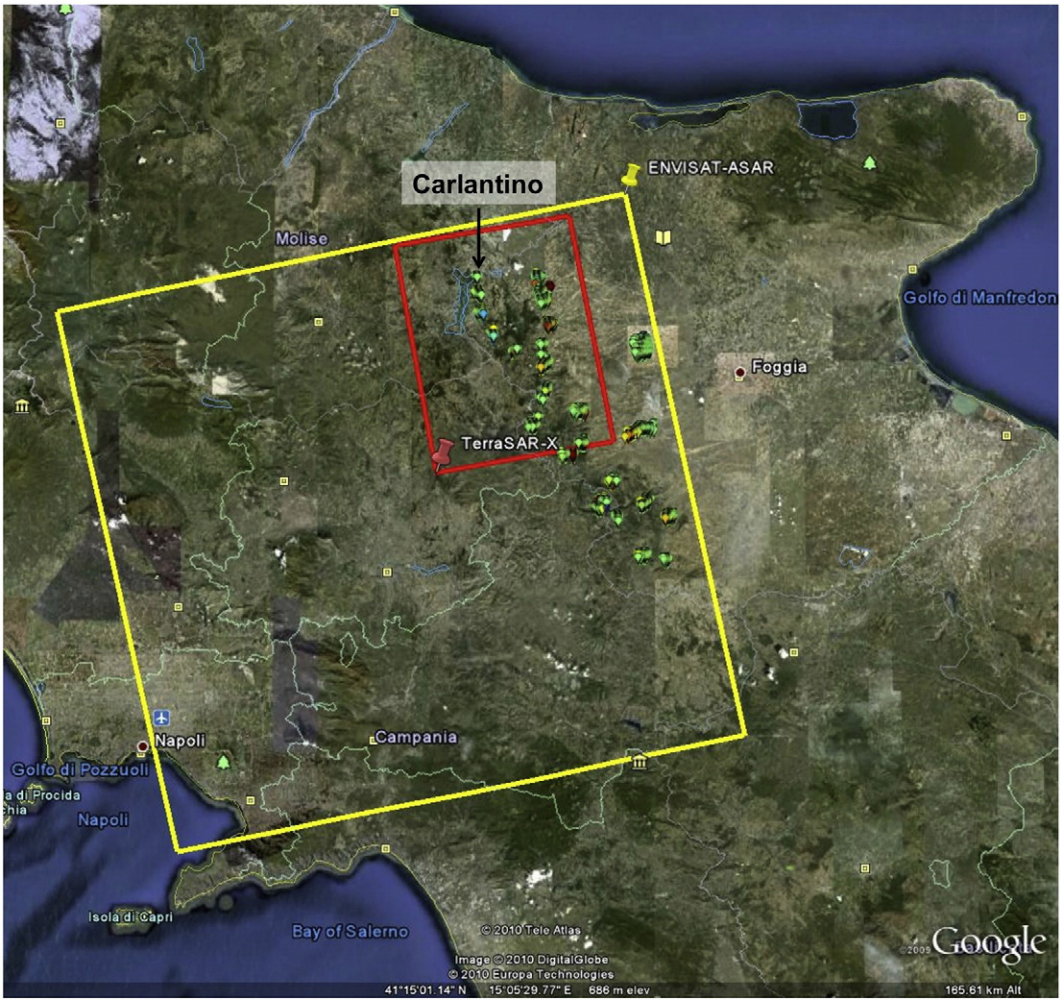


Fig. 6. Yellow and red frames correspond to the ground coverage in Daunia Region of the ENVISAT ASAR and TSX images, respectively.

20 cm deep; CRs were then fixed to the basements using bolts (Bovenga et al., 2012b). By regulating the bolts, CRs can be slightly re-oriented, in order to appear visible in acquisitions with moderately different incident angles.

A relatively small AR size was preferred, to minimize accidental bending of the side panels, the effects of wind, the exposition to vandalism, and to allow easier transportation and deployment in the typical harsh terrain settings encountered on the ground. Thanks to the short X-band wavelength, as well as to the small resolution cell of TSX data, small-sized CRs still provide a sufficient radar cross section to be visible in the images.

The CR locations were planned considering the tradeoff between the constraints for designing a geodetic network as dense as possible, and those imposed by conditions of CR visibility and ground accessibility. Fig. 7 shows in white the CR locations superimposed on an optical image from Google Earth. The harsh in situ conditions of the site strongly limited the possible CR locations, as only 6 CRs could be installed following the above-mentioned guidelines. In particular, the distance limit between CRs, needed for a reliable atmospheric signal suppression (as detailed in Section 3.2), could not be strictly respected for most CR pairs. The presence of trees and walls (in particular on the top of the landslide), made problematic to completely fulfill the geometric deployment conditions needed to avoid the visibility problems sketched in Fig. 8(*b*) (namely shadow and layover areas). Furthermore, the land cover often resulted not ideal, because of the presence of vegetation or scattered debris, causing strong radar clutter, which negatively affects the CR RCS.

In order to properly integrate the InSAR phase measurements coming from PSI and ARs, CR-1 was deployed on the top of the landslide, close to structures (walls and guard rails) known to behave as PS in the pre-existing X-band SAR stack (see Fig. 7). In this case, the constraints needed to avoid interaction with SAR echoes backscattered by the walls behind and in front of the CR were particularly severe. CR-3 was, instead, installed on the landslide body, thus requiring an ad hoc setting: in order to make the CR-3 motion integral with the sliding body, a 1.5-m long iron bar was staked into the ground and then fixed to the basement (Bovenga et al., 2012b).

3.2. Analysis of CR SAR amplitude stability

After CR deployment, performed on early July 2011, 26 TSX images were acquired between 12th July 2011 and 30th December 2013. The CRs were identified on these SAR images by applying a threshold to the intensity, followed by an interpolation step needed to determine

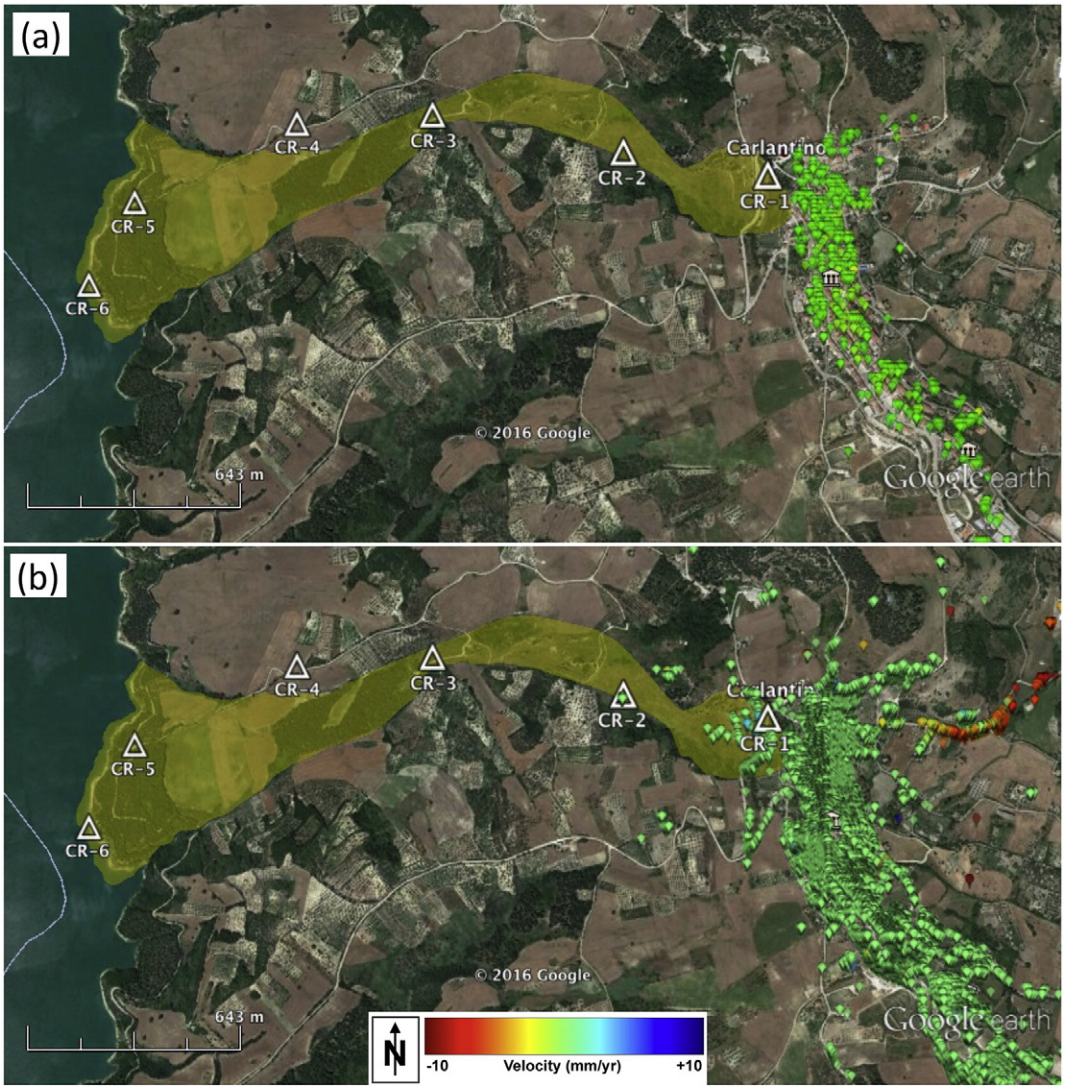


Fig. 7. PSI results on the Carlantino site. The colored dots are the PS identified in the area by processing ENVISAT ASAR (a) and TerraSAR-X data (b), respectively. CR locations are sketched as withe triangle. The colored area corresponds to the "Rione Toppo" landslide polygon derived from Canora et al. (2012).

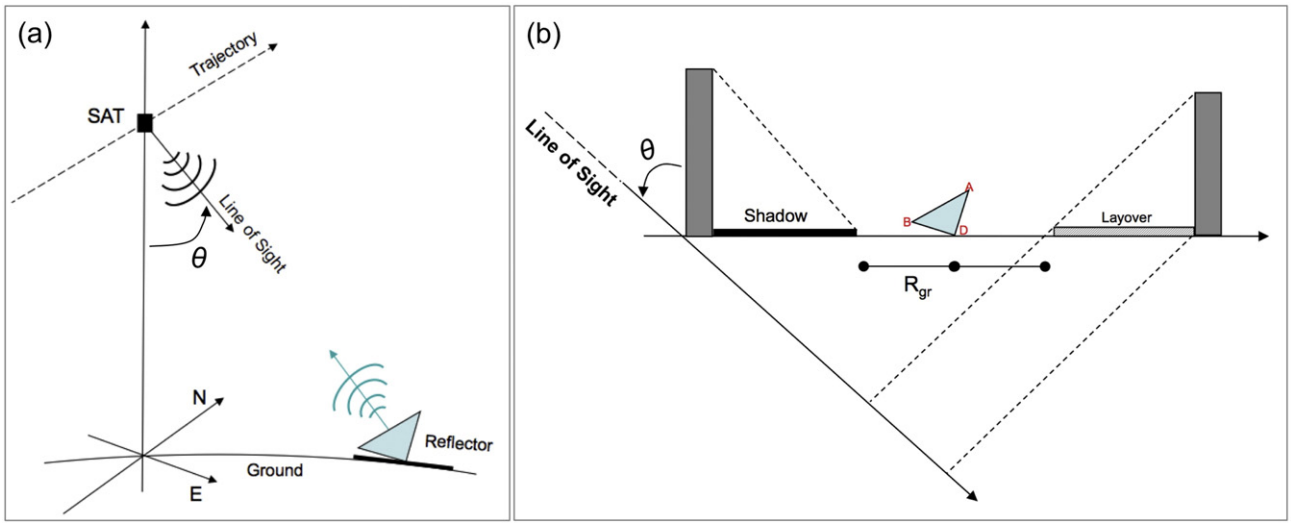


Fig. 8. (a) Geometrical requirements for the artificial reflector visibility from the satellite. (b) Sketch of SAR acquisition over an artificial reflector.

their sub-pixel locations. For all the CRs, the backscattered radar signal dominates the background, showing the typical *sinc* profile that can be easily identified on the intensity image. The signal phase follows the expected behavior as well, being constant within the main lobe of the amplitude profile.

In order to verify the quality of CR backscatter, we performed an absolute radiometric calibration of the SAR images, by applying corrections for a constant term and for incident angle variations. The resultant intensity values computed for each CR at the different acquisition times, approximate the normalised RCS, and were used to investigate the CR backscatter stability. Neither CR-5, nor CR-6 could be properly identified in all the images of the SAR intensity stack. As it turned out, they were subject to vandalism early after deployment, thus being unusable for the interferometric analysis. For the other CRs, the calibrated intensity values are quite stable in time, ranging between 14 and 18 dB (see Fig. 10). These values are in some cases slightly lower than those theoretically predicted, probably due to both the cheap manufacturing and the interaction with the clutter. On three scenes, acquired on January and August 2013, a strong RCS decrease was found to affect all the CRs. We attribute this to possibly inaccurate values for the calibration constants for these acquisitions, although no further evidence of this effect could be found. Anyway, these three acquisitions were not used in the InSAR phase analysis described in the following. The stability index (SI) was also computed for each CR, as the ratio between the mean and the standard deviation (STD), of the RCS estimated along the temporal axis. The SI values are listed in the legend of Fig. 10. For stable CRs, the SI values are comparable with those obtained in other studies using similar CR size and SAR data (e.g. Yu et al., 2013). CR-6 shows clearly a very poor stability index, due to the mentioned vandalism exposure, which induced strong tilting of the CR structure, thus destroying the CR alignment with the sensor LOS. Also CR-1 shows low SI value, which is unexpected for its location, but in line with the results from the InSAR phase analysis described in the following.

3.3. Processing of CR InSAR phase

The CR amplitude analysis was followed by the InSAR phase analysis, devoted to derive precise displacement measurements. The InSAR phase is obtained by coupling two SAR images acquired during successive satellite passes. It is proportional to the difference between the two sensor-target distances, which depends on: i) the target elevation with respect to a reference surface; ii) the target displacement occurred along the LOS between the two satellite passes; iii) the refractivity index changes due to the presence of the atmosphere; iv) decorrelation noise (Hanssen, 2001).

According to this model, the InSAR-based displacement measurement requires the following steps: selecting ground targets behaving

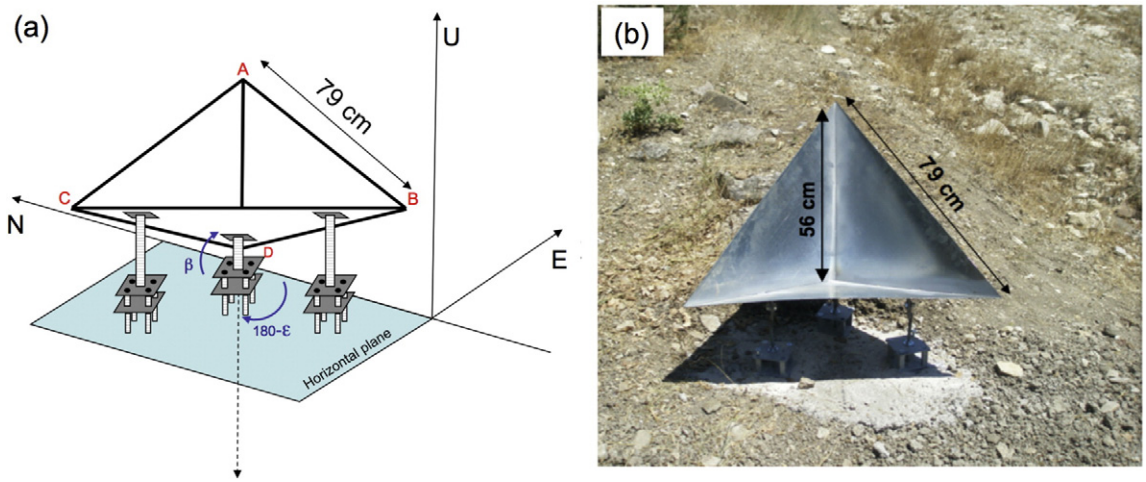


Fig. 9. Trihedral corner reflector scheme (left) and picture (right).

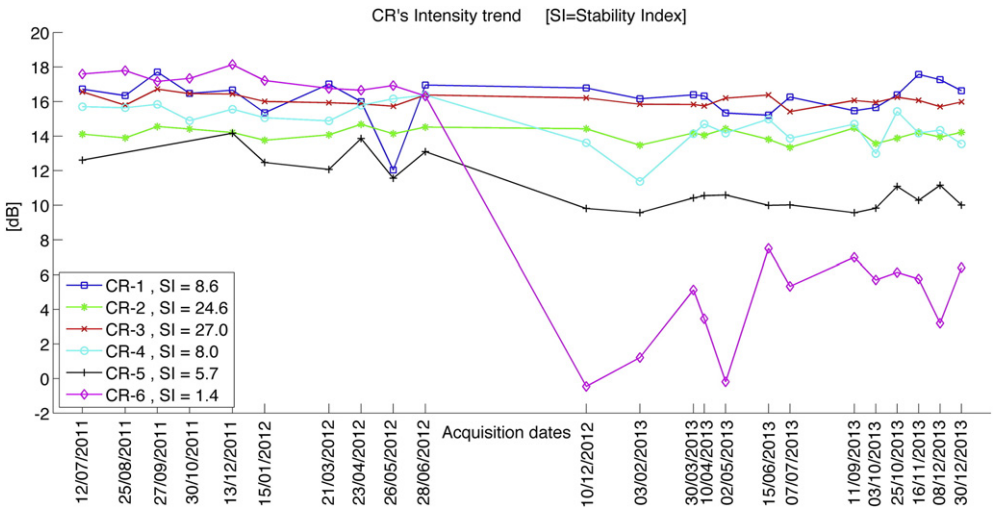


Fig. 10. Trends of the CR's intensity values computed for each SAR acquisition by performing radiometric calibration. The Stability Index values are listed in the figure legend.

coherently in time (i.e. for which decorrelation noise is negligible); removing the topographic phase contribution by using an independent elevation model; identifying and attenuating the atmospheric disturbances. These steps are performed by multi-temporal InSAR algorithms (e.g. PSI), and rely on processing of long temporal series of SAR data. The atmospheric signal estimation is performed thanks to its stochastic behavior, which is known to be totally uncorrelated in time, since SAR images are acquired at intervals of several days, but correlated in space (Hanssen, 2001). The displacement estimation reliability is influenced by both the number of SAR images, and the number and the spatial distribution of coherent targets, either natural (PS) or artificial (CR). In particular, a low number of coherent targets, or their non-uniform distribution over the radar image, can hamper a reliable estimation of atmospheric signal (Refice et al., 2011) and/or displacements field.

According to this, and having available just a few CRs, we adopted a processing scheme, detailed in Bovenga et al. (2014), sketched in Fig. 11, and summarised in the following:

Step -1: a first displacement map is derived by running standard PSI processing chain over an area that includes the Carlantino village and the surrounding slopes (including the "Rione-Toppo" landslide).

Step -2: the phase analysis over the CRs is refined by combining the PS locations and displacements derived from the PSI processing, and the accurate differential GPS measurements of the CR's positions.

An InSAR phase stack was obtained by following the single master approach adopted by the SPINUA algorithm (Bovenga et al., 2004), which was used to perform the subsequent PSI processing. Then, we derived a first PS deformation map over the area of interest by processing through the standard SPINUA chain the TSX scenes acquired after CR deployment (from July 2011 up to December 2013).

The map in Fig. 7(*b*) shows that CR-2 was properly detected as a PS in this InSAR stack, thanks also to the presence of other natural PS nearby, which allows reliable estimation and compensation of the atmospheric phase component. Although CR-3 and CR-4 have good amplitude stability, they were not detected as PS stable in phase, probably because they lie too far away from other useful natural PS, so that removal of spatially variable phase contributions was unreliable at their locations. Also, CR-1 was not recognized as a PS, probably because the area around was surprisingly used to dispose waste material, including large metal objects, such as washing machines. Moreover, we cannot exclude that the CR was hit during the waste disposal operations, possibly causing shifts of its phase center, and thus corrupting the InSAR phase content. This prevented using CR-1 for defining a reference deformation value for the InSAR phase analysis specific for CRs (Step 2), as it was planned during the deployment phase. Instead, this role was assumed by CR-2, whose velocity was computed through PSI in the Step 1 of our procedure, having as reference the stable area on the hilltop, and found to be negligible. This is not surprising, since CR-2 lays on the border of the landslide body, in its upper part, which was subjected to consolidation works, thus it is reasonable to expect that the target is stable.

According to this, in the Step 2, the estimation of the displacements occurring on the other usable CRs was performed with respect to CR-2, by exploiting the differences between the InSAR phase values of neighbouring CRs, and hypothesizing a limited impact of atmospheric disturbances.

Fig. 12 shows two differential InSAR phase fields computed on the landslide area by coupling acquisitions separated by 11 days, which

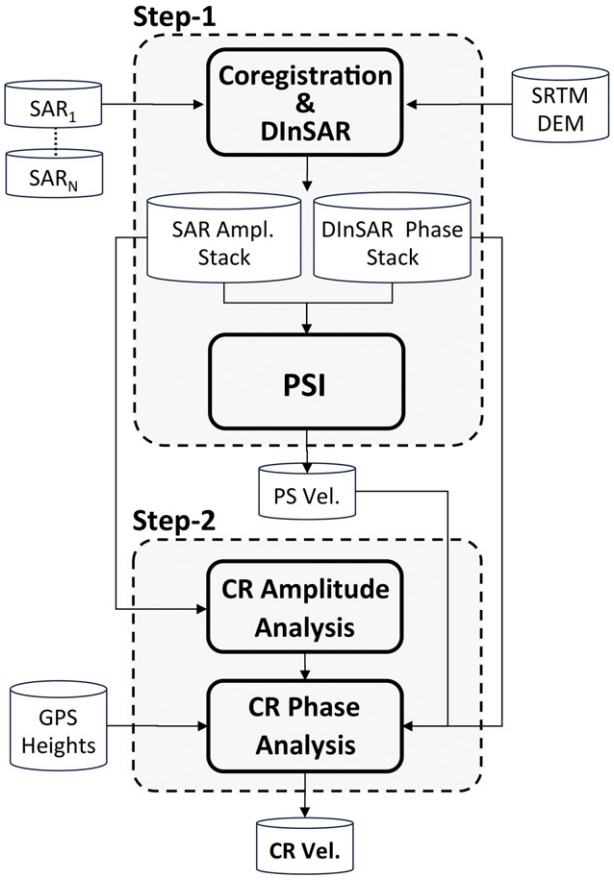


Fig. 11. Processing flow chart.

show generally low average temporal decorrelation. Thanks to the high InSAR coherence, the spatial distribution of the phase is visible, showing a pattern that is basically made by two components: 1) local scale variations related to topographic features, missing in the low resolution digital elevation model derived from the Shuttle Radar Topography Mission (SRTM) mission (Rabus et al., 2003), which is commonly adopted in standard InSAR processing; 2) bubble-like features typical of tropospheric turbulence structures. The use, in Step 2, of the precise GPS heights for the CR analysis, limits the impact of topographic residuals in the velocity estimation. The bubble-like features, instead, typical of tropospheric turbulence structures, suggest the occurrence in the area of considerable atmospheric phase contributions, probably due to convective motions induced by the sun heating the reservoir water downslope. According to other studies (e.g. Crosetto et al., 2013), the distance between CRs should be limited to a few hundreds of meters, to guarantee negligible atmospheric effects at least in the estimation of the deformation between CRs. Due to the above-mentioned limits, the usable network is reduced in our case to only 4 CRs, and only three connections between CRs have a distance well below 500 m (namely those between CR-1 and CR-2, CR-3 and CR-4, and CR-5 and CR-6). This means that, despite the careful planning adopted, residual atmospheric noise can still affect the phase values.

In view of this, to limit unmodeled differential InSAR phase variations, the topographic phase component estimation was refined, for each CR, by using independent differential GPS measurements of centimetric accuracy, instead of the low resolution SRTM digital elevation model. These differential height values derived from the GPS measurements were used to constrain the estimation of the differential displacement velocity between two neighbouring CRs ($\Delta V_{i,j} = V_j - V_i$).

Finally, in order to check the reliability of the estimations, we computed the sum of the estimated differential phase values along closed paths (namely, the circuitation) defined by triangulations between the CRs, as sketched in Fig. 13. Where the circuitation is non-null, the estimation is not reliable due to phase aliasing, and should be dismissed.

4. Results and discussion

According to the processing scheme described in the previous section, we derived the mean displacement rate and the time series for each CR's couple sketched in the Fig. 13. The standard deviation of the phase residues with respect to the estimated trend is higher for the estimations involving CR-1 than for the others, thus indicating the noisy behavior of its InSAR phase, that confirms the fact that it shows low amplitude stability, and was not recognized as a PS by standard PSI (Step 1). Moreover, while the circuitation computed between CR-2, CR-3 and CR-4 amounts to only −0.1 mm/y, well within measurement uncertainties, the circuitation that involves CR-1 yields the non-negligible value of −7.4 mm/y, thus confirming the low reliability of the InSAR phase of this CR, and the possible occurrence of unwrapping errors in

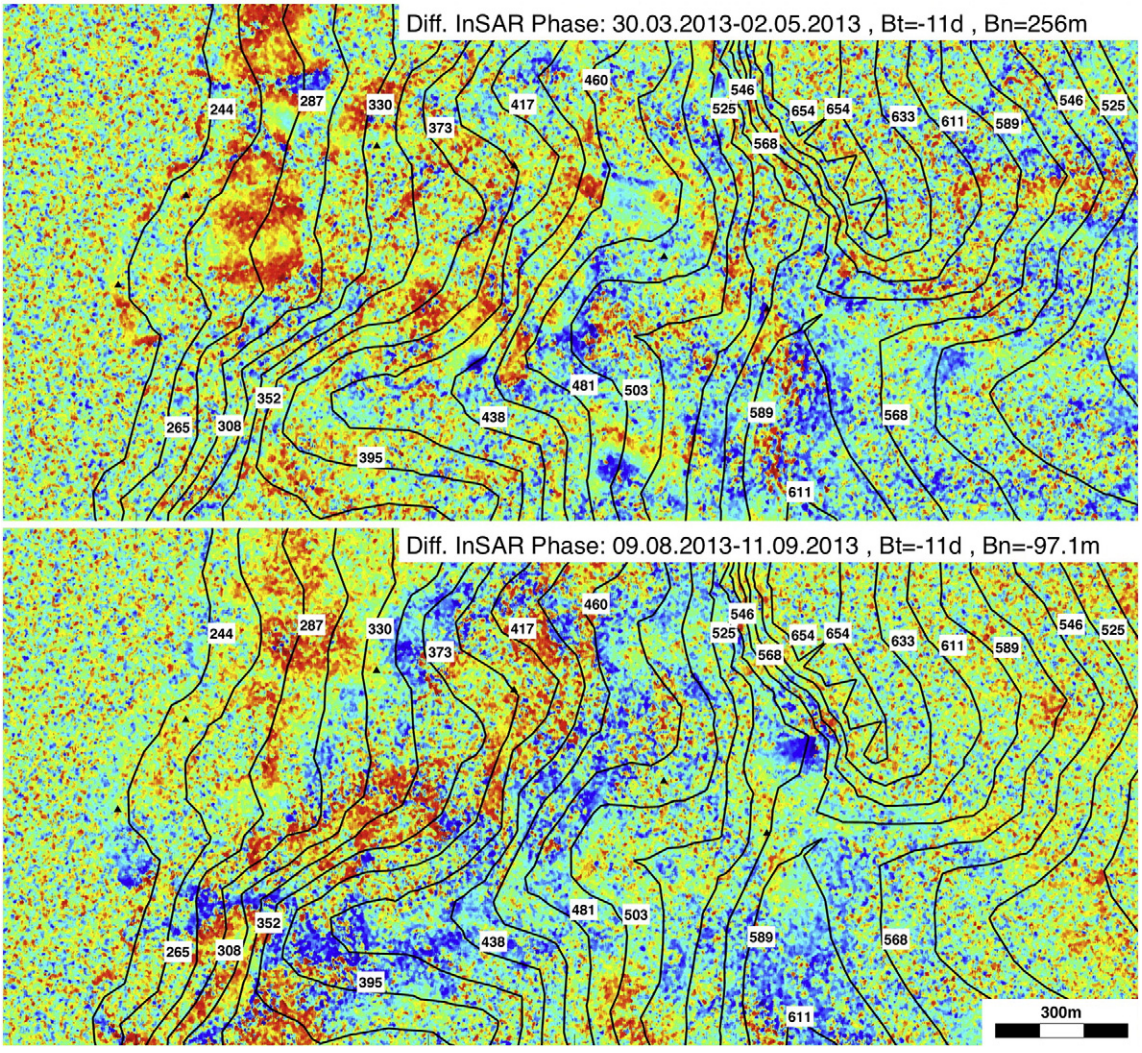


Fig. 12. Differential phase fields obtained by coupling adjacent acquisitions. Contour lines derived from SRTD MED are superimposed in black with labels reporting the elevation in meters. CR locations are sketched as black triangles.

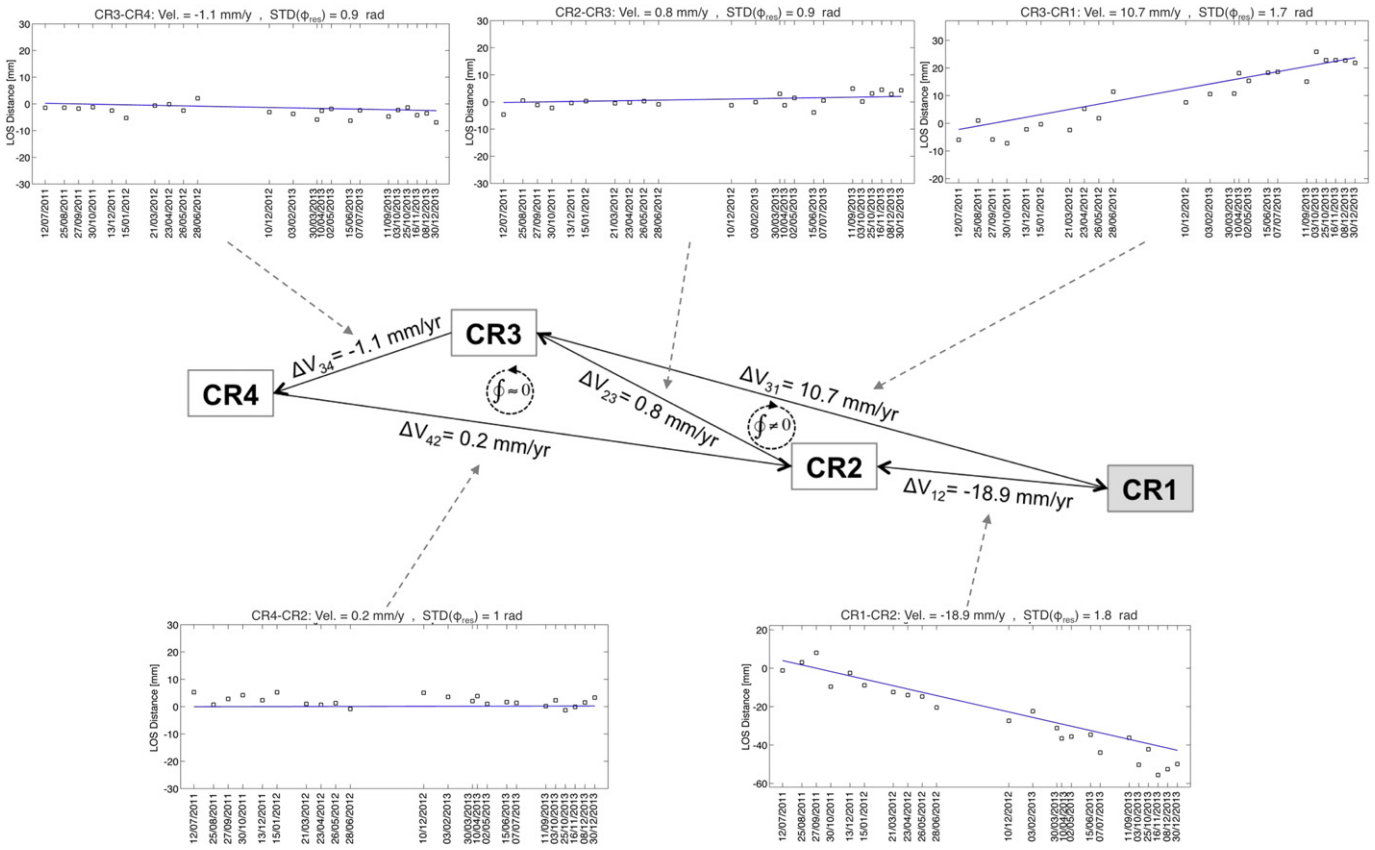


Fig. 13. Sketch of the connections between CRs used for the displacement estimation, and the relative time series and linear trends. Mean displacement rate and standard deviation of the phase residuals are reported on each plot. The circuitation that involves CR-1, already recognized as unreliable, is non-null.

the displacement estimation. Consequently, CR-1 has been dismissed from the final displacement analysis.

Taking into account these results, we were able to measure only the differential velocities involving CR-2, CR-3, and CR-4 (see Fig. 13). In particular, the estimated differential velocities ΔV_{23} and ΔV_{34} show negligible values. According to these values, and to the CR-2 velocity derived from PSI processing ($V_2 = V_{PS}$), the velocity of CR-3 and CR-4 can be computed as follows: $V_j = V_{j-1} + \Delta V_{j-1,j}$ with $j = 3, 4$. Since, from the PSI analysis, CR-2 is stable ($V_2 = 0$ mm/y), the final CR velocities are V3 = 0.8 mm/y and V4 = −0.3 mm/y. These values are below the noise floor affecting the measurement, and consequently can be considered negligible, proving the stability of the ground area

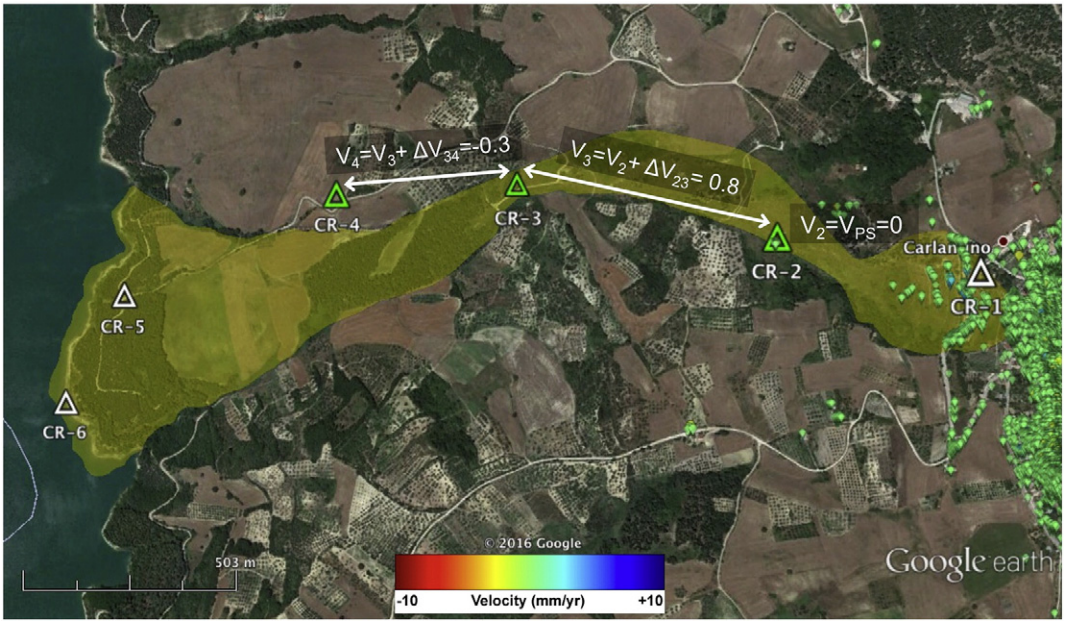


Fig. 14. CR displacement rates in mm/y computed according to the double difference DInSAR analysis and PSI results. Reliable CRs are colored according to the color bar used for the PSI velocity map. Unreliable CRs are in white.

around these CRs. Fig. 14 shows the CR displacement rates in mm/y computed according to the double difference InSAR analysis described above. Reliable CRs are colored according to the color bar used for the PSI velocity map, while unreliable ones are in white. The landslide is marked by a yellow-filled polygon.

The main points of interest of our investigation, related to the behavior of the landslide are the large extension of the landslide body and, according to this, the possible coexistence in the upper, lower and lateral parts of independent strain and stress fields. At the toe of the landslide, a large accumulation of debris occupies and diverts the previous Fortore riverbed, now submerged by the water of an artificial reservoir. Evidences of small displacements of this part of the landslide body arise from the cutting of inclinometers tubes placed in the area, at different times. The distance of the landslide toe from its crown, over 2 km, does not allow the calcareous debris of the crown to reach the accumulation zone at the toe of the slope. As common in longitudinally very developed landslides, the stress field along the channel can be substantially different in its parts, compressive beneath the feeding zone and above the accumulation zone at the toe, extensile along the middle part in the channel.

Our analysis, thanks to the use of CRs and the described processing strategies, allows to propagate downslope the measurements obtained through standard space-borne InSAR PSI analysis, proving the stability of part of the landslide body (Fig. 14). These results are coherent with those deriving from traditional monitoring. In particular, in can be inferred that, forty years after the last catastrophic movement of the upper part of the landslide, occurred in 1976, the accumulation zone at the toe has not yet been reached by the dislocated mass. The last contribution of debris to the landslide body comes from rocky members of the flysch formations; they drain more easily and have much friction to move downwards.

Therefore, the results from InSAR and CRs analysis were able to prove the stability of the upper and central part of the landslide body and, consequently, the effectiveness of the different and subsequent works carried out for adjusting and consolidating the landslide area. Moreover, since InSAR results confirms the stability of the central and upper part of the landslide, the small deformations still active at the toe (as from in-situ measurements) seem to respond to other kinds of stress, such as e.g. the seasonal level variations of the reservoir.

5. Conclusions

This work provides an example of InSAR application in a really complex setting, quite common in areas affected by slope instabilities in rural or peri-urban areas where coherent targets are lacking: the “Rione Toppo” large landslide (southern Italy) located in unfavorable settings for InSAR applications. Despite several stabilization works, movements on the toe of the landslide were detected through inclinometers.

The current activity state of landslide has been thus investigated by using satellite multi-temporal InSAR analysis of X-band, high-resolution TerraSAR-X data, and a network of passive CRs to overcome the lack of coherent targets on the main landslide body. Although reflectors could be theoretically deployed to fulfill all the requirements of a reliable geodetic network, realistic in situ conditions often hinder ideal settings, as in the case of the Carlantino landslide. Moreover, two out of the six CRs were subjected by vandalism after deployment, so that the number of usable CRs for our experiment further decreased.

Despite the non-optimal distance between CRs, a null mean relative displacement rate was reliably estimated for all the usable CRs by exploiting the differences between the InSAR phase values of neighbouring CRs, the PSI results, and the accurate CR positions derived from differential GPS.

We can conclude that the use of CRs allowed propagating the PSI results downslope, proving the stability of the upper and central part of the landslide body, and relating the main cause of landslide activity at the toe, to the reservoir water level seasonal variations.

A series of lessons can be learned from the described experiment, which suggest some recommendations and guidelines to follow in designing and deploying CR networks on difficult landslide sites:

- unlike devices used for sensor absolute calibration, CRs installed with the purpose of displacement monitoring do not pose strict constraints in terms of correct manufacturing, allowing cheap solutions to be adopted, especially for use with short-wavelength SAR sensors;
- deployment is, instead, the most critical phase, requiring great care in planning CR positions in order to guarantee visibility of each individual CR and consistence of the entire CR network; factors to be considered in the planning should include environmental and climatic settings, geometric line-of-sight constraints, land cover analysis, local present and foreseen conditions of access routes to planned locations, possible legal or insurance constraints on the deployment. Since often conditions on the ground, or even post-deployment events, may cause disruptions of one or more nodes of the CR network, it is recommended to include a consistent degree of redundancy in planning deployment positions to render it more robust to failures; for instance, in the present case only three CRs out of the six deployed were finally employed for the surface displacement analysis, which means a 50% success rate;
- to improve data analysis, ancillary data acquisition, such as precise positions via GPS, in situ temperature, pressure, and general weather records, are recommended. Periodic visits to secure CR function and integrity should be foreseen for the duration of the experiment;
- finally, specific processing solutions for satellite data should be adopted, in case external factors hinder or forbid successful processing through standard procedures.

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